

GAR, Russian Federation

WMO: 313840

Lat: 52.567N

Lon: 129.067E

Elev: 246

StdP: 98.40

Time Zone: 9.00 (E09)

Period: 86-95

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -42.5	(c) -40.6	(d) -44.3	(e) 0.0	(f) -42.4	(g) -42.2	(h) 0.1	(i) -40.4	(j) 6.8	(k) -18.5	(l) 4.9	(m) -19.1	(n) 0.0	(o) N/A	(p) 0.599

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	13.0	28.7	20.2	27.4	20.0	25.7	18.8	22.5	26.0	21.2	25.0	20.2	24.1	1.5	N/A

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.4	16.5	24.3	20.1	15.2	23.0	18.9	14.2	21.9	67.6	26.2	63.3	25.0	59.7	24.1	29.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a) 6.3	(b) 4.9	(c) 3.5	(e) -45.3	(f) 32.5	(g) 2.7	(h) 1.9	(i) -47.2	(j) 33.8	(k) -48.8	(l) 34.9	(m) -50.3	(n) 35.9	(o) -52.3	(p) 37.3		
(a) 6.3	(b) 4.9	(c) 3.5	(e) -44.7	(f) 25.4	(g) 2.0	(h) 2.3	(i) -46.2	(j) 27.0	(k) -47.4	(l) 28.4	(m) -48.6	(n) 29.7	(o) -50.1	(p) 31.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	-3.0	-28.6	-21.3	-11.5	0.4	8.6	16.3	18.5	16.0	8.5	-0.4	-16.8	-26.7		
	DBStd	17.22	5.00	5.03	6.15	4.74	4.43	3.02	2.73	3.10	3.89	4.82	5.84	5.25		
	HDD10.0	5452	1197	876	666	289	82	1	0	1	74	324	805	1137		
	HDD18.3	7851	1456	1109	924	538	303	74	33	86	294	582	1055	1396		
	CDD10.0	706	0	0	0	0	37	191	262	186	30	0	0	0		
	CDD18.3	64	0	0	0	0	0	14	37	13	1	0	0	0		
	CDH23.3	1009	0	0	0	0	48	305	457	195	4	0	0	0		
	CDH26.7	204	0	0	0	0	8	64	96	35	1	0	0	0		
(14) Wind	WSAvg	0.9	0.4	0.6	1.0	1.6	1.6	1.1	0.9	0.9	0.9	0.9	0.7	0.3		
(15) Precipitation	PrecAvg	563	7	6	11	23	62	78	133	119	65	30	19	10		
	PrecMax	797	24	16	45	59	135	137	273	203	122	74	46	23		
	PrecMin	410	1	0	0	3	15	36	43	33	27	6	3	2		
	PrecStd	107	5	5	9	14	33	29	66	45	26	15	11	6		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	-10.6	-3.5	10.5	17.1	27.9	30.5	30.4	29.6	23.6	15.7	0.4	-9.9		
		MCWB	-12.0	-5.5	5.1	8.6	16.8	19.4	21.4	23.1	16.7	8.1	-1.7	-10.3		
	2%	DB	-14.3	-6.2	4.8	14.7	24.1	28.4	28.8	27.1	21.5	13.1	-2.4	-12.4		
		MCWB	-14.8	-7.9	0.6	6.9	14.1	18.9	21.2	21.1	15.1	6.9	-4.0	-12.9		
	5%	DB	-17.2	-9.1	1.9	11.8	21.4	26.8	27.7	25.4	19.4	10.8	-4.0	-15.0		
		MCWB	-17.7	-10.6	-1.4	4.8	12.4	18.5	20.9	19.1	14.5	5.3	-5.3	-15.5		
	10%	DB	-19.3	-11.5	-0.6	9.5	18.6	25.0	26.0	23.6	17.0	8.5	-6.3	-17.4		
		MCWB	-19.8	-12.7	-3.4	3.4	10.5	17.6	20.1	18.4	12.6	4.4	-7.5	-17.8		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-11.6	-5.0	5.1	9.2	16.8	21.8	25.6	25.1	18.6	9.3	-1.4	-10.2		
		MCDB	-11.0	-3.7	10.7	16.1	25.1	27.0	28.0	28.3	21.5	14.1	0.2	-9.8		
	2%	WB	-14.7	-7.8	1.1	6.9	14.6	20.9	23.1	21.7	16.7	7.5	-3.7	-12.8		
		MCDB	-14.2	-6.2	4.0	13.6	21.8	26.0	26.4	25.3	19.9	11.4	-2.5	-12.4		
	5%	WB	-17.7	-10.5	-1.3	5.4	13.2	20.0	21.9	20.3	15.0	5.9	-5.4	-15.5		
		MCDB	-17.2	-9.1	1.7	11.0	20.5	24.4	25.3	23.7	17.7	10.3	-4.1	-15.0		
	10%	WB	-19.7	-12.6	-3.1	3.8	11.4	18.9	20.9	19.1	13.3	4.4	-7.6	-17.8		
		MCDB	-19.3	-11.4	-0.9	8.8	17.4	22.9	24.6	22.7	16.6	8.1	-6.4	-17.4		
(35) Mean Daily Temperature Range	5% DB	MDBR	16.0	18.8	17.4	13.7	14.9	14.6	13.0	13.2	14.3	14.2	14.2	14.4		
		MCDBR	15.5	20.1	18.3	17.1	20.5	19.3	16.4	16.7	17.4	18.4	13.8	14.7		
		MCWBR	14.9	18.3	14.2	11.2	11.8	10.3	9.6	10.3	11.7	12.4	12.2	14.1		
	5% WB	MCDBR	15.6	20.0	17.4	15.9	18.7	14.2	13.4	14.3	15.1	16.9	12.9	14.7		
		MCWBR	15.0	18.2	13.6	11.0	11.0	8.9	9.7	10.0	11.3	12.1	11.8	14.0		
(40) Clear-Sky Solar Irradiance	taub		0.255	0.289	0.355	0.536	0.587	0.500	0.474	0.418	0.388	0.395	0.311	0.256		
	taud		2.146	2.084	1.952	1.691	1.686	1.998	2.125	2.276	2.273	2.056	2.093	2.096		
	Ebn at Noon		730	806	809	692	683	762	777	804	781	657	625	641		
	Edh at Noon		69	101	143	214	226	167	145	118	105	105	74	60		
(44) All-Sky Solar Radiation	RadAvg		1.39	2.58	4.29	5.14	5.33	6.06	5.42	4.71	3.57	2.33	1.47	1.07		
	RadStd		0.05	0.07	0.18	0.51	0.39	0.58	0.49	0.31	0.28	0.17	0.08	0.05		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
Regional (8 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page